An Autonomous Robot for Delivery Services

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Abstract:

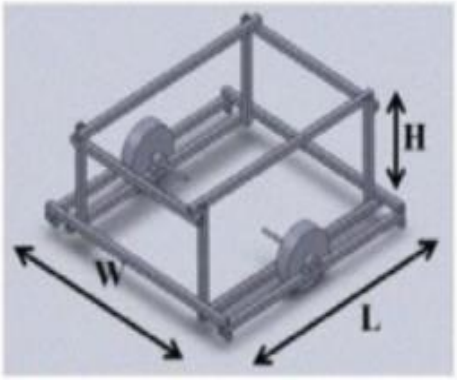
The medical industry could be transformed by delivery robots. It aids in reducing the risk of human error by increasing efficiency and accuracy. One of the primary advantages of delivery robots is that they can reduce the time it takes to supply medication to patients. This is especially important when medications or blood samples must be transported quickly and efficiently. These self-driving robots can also help to reduce the risk of infection transmission by reducing the need for human interaction with patients. This is especially important during pandemics like covid, when it plays a critical role in limiting virus spread. There are several challenges associated with the use of delivery robots in the medical industry, including the need for robust navigation systems to ensure that the robots can navigate hospital environments safely, and a secure storage compartments to prevent medical supply theft or tampering, and effective communication systems to ensure that the robots can be controlled and monitored by medical staff and further implementations will be added on in future.

Keywords:

Autonomous Robot, Machine Learning (Image Processing), Self-Driven Robot, Robot Logistic, Artificial Intelligence.

Introduction:

Delivery robots are autonomous machines that are designed to transport goods from one location to another in hospitals. These automated robots can navigate complex environments and use a variety of sensors and cameras to detect obstacles and avoid collisions with the environment. As the technology continues to evolve, we can expect to see even more advanced delivery robots that are better equipped to meet the specific needs of different industries. The use of delivery robots is expected to continue to grow in the coming years, as companies and organizations seek to improve the efficiency and sustainability of their delivery operations. In order to implement this idea few papers have been reviewed.



Isotopic view of the motion system

A. Paper 1 Summary:

In the first paper the product is being exposed to the environment so that theft or tampering may occur. In order to rectify this issue the robot will be provided with a password protected case for security purposes. The proposed robot prototype has been designed for the accomplishment of four task. They are carry the package securely in a password protected container, navigate through the GPS waypoint and reach the destination successfully, delivery of the package is done without any human contact, return back to the home location. Drawbacks are lack of accuracy in the direction of Google map. To accommodate more packages and heavy load as it can't carry more than a kilogram, the size of the robot can be increased and powerful motors should be attached.

B. Paper 2 Summary:

In the second paper the robot is being used to serve the patients by taking the prescription to the pharmacy and by collecting the samples like blood and x-ray and etc., and also used to avoid physical contact with a healthy person and an infected person. In this case an automation robot is really helpful. Thus, this research aims at designing an Automated medical Delivery System to overcome the above mentioned issues with the current systems. The proposed robot prototype has been designed for the accomplishment of four task. They are the delivery is done by connecting colored lines which link table and kitchen for guiding track, signals are sent to the robot and the dishes are placed in a tray by the chef, delivery system will give signal to ordering system for delivery confirmation, return back to their standard mode after delivery. The drawbacks are this system can't be used for large number of tables as guiding track for the robot are made using connecting lines and the food placed in the tray will be prone to dust.

C. Paper 3 Summary:

In the third paper A robot with known destinations, a toll system that opens lock and delivers packages ,a robot whose movements are initiated by a start and stop signal. GSM (Global System for Mobile communication) Navigation method is implemented. The proposed robot prototype has been for the accomplishment of four task. They are autonomous robots with predetermined destinations, GSM (Global System for Mobile communication) Navigation method is implemented, in response to start button it moves and stops when stop signal is encountered. OTP received by the user and entered in the keyboard of robot, the lock is opened and delivered.

D. Paper 4 Summary:

In the fourth paper SLAM generates the maps, input commands and robot information are displayed on the Microsoft Surface Pro 5, laser pulses are sent to the surface/target to measure their return time, and a real sense viewer lets the robot stay on track. The proposed robot

prototype has been designed for the accomplishment of four task. They are the mapping can be generated automatically by SLAM, input commands and Robot information were displayed on the Microsoft surface pro 5,laser pulses are sent to surface/target to measure their return time, real sense viewer allows the robot to avoid obstacles by itself. Drawbacks of the project are a BOT can travel at a speed of 0.2m/s for two hours and there is a lack of proper implementation of the security system.

Frame work:

Robot Operating System(ROS) is an open-source framework that provides a collection of tools, libraries, and conventions that can be used to develop complex robot applications. ROS is widely used in automation robots to provide a standardized platform for communication and control between various components of a robot system. It uses a node-based architecture, where different components of a robot system are implemented as nodes that communicate with each other using ROS messages. Different types of nodes are used to transfer information such as control nodes, communication nodes, sensor nodes and navigation nodes. For example, We use sensor nodes to capture data from sensors such as cameras, ultrasonic sensors. These nodes can be used to process the data and publish it to other nodes in the system by using ROS. Using this node-based architecture, ROS allows us to create complex robot systems by composing a set of individual nodes that communicate with each other.

Algorithms:

1.A* algorithm:

The A* algorithm is a path finding algorithm that is commonly used in automation robots to find the shortest path between two points in a given environment. It is a widely used algorithm because it is efficient and can find the optimal path quickly, even in complex and dynamic environments.

$$F(n) = G(n) + H(n)$$

Where,

F(n) -> estimated total cost of reaching the destination from the starting node through node n

G(n) -> cost of reaching node n from the starting node

H(n) -> estimated cost of reaching the destination from node n based on a heuristic function. The distance between two points is found using Euclidean distance. The Euclidean distance between two points in a 2D plane can be calculated as follows:

$$d = \text{sqrt}((x2 - x1)^2 + (y2 - y1)^2)$$

Where,

d is the distance between the two points (x1, y1) and (x2, y2) are their respective coordinates.

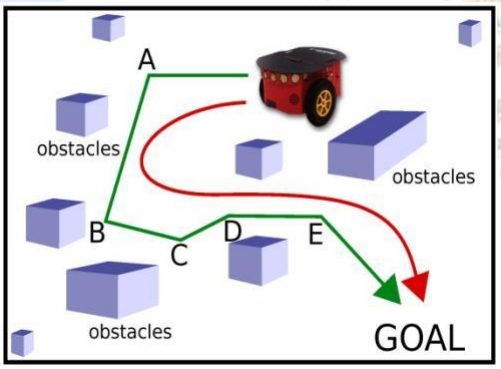


Fig 1:Pathway detect by A* Algorithm

Block Diagram:

The robot gets the information from the instructor. The Robot Operating System(ROS) is used as the framework for the robot. Ultrasonic sensors are used as the obstacle detector for the robot. The robot also uses image processing to identify the people in the surroundings.

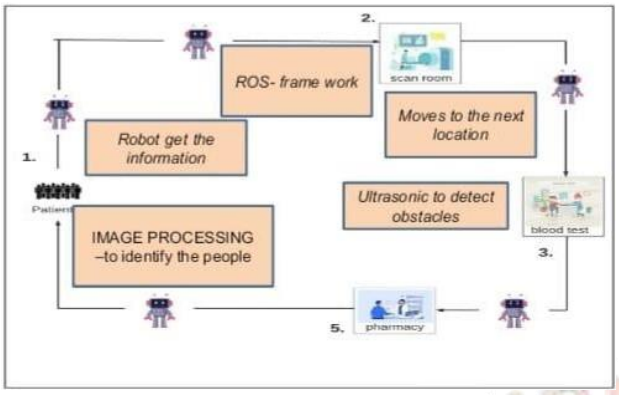


Fig 2: Block Diagram

Mind Map:

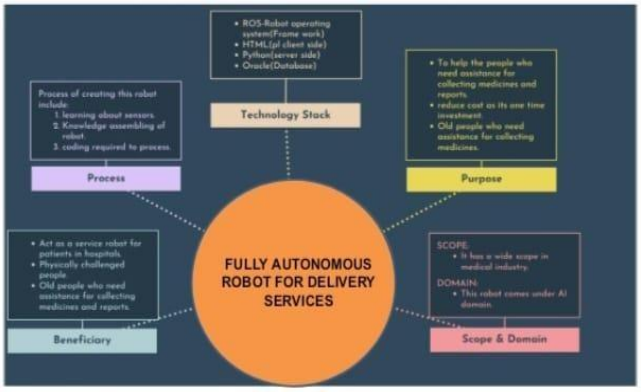


Fig 3: Mind Map

Requirements Specification:

A.Hardware Components:

(a)Raspberry Pi:

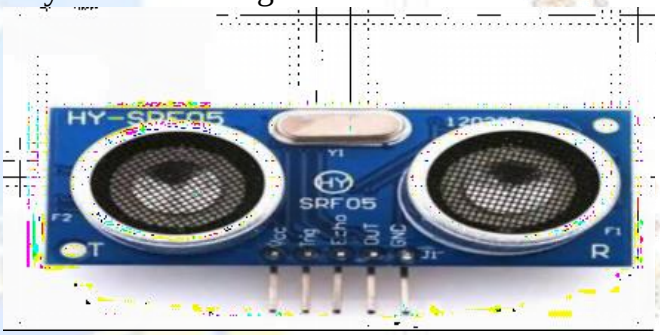
The Raspberry Pi board is the perfect computer to embed in our robot. The footprint is very small, credit card size and it's powerful enough to run some advanced programs, algorithms, and frameworks, for example robot operating system (ROS).



Raspberry Pi

(b) Ultrasonic Sensor:

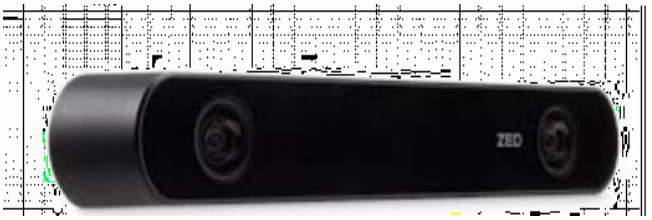
Ultrasonic sensors are utilized to detect the large obstacle and to obtain information between the robot and obstacle. Ultrasonic sensors can be used to detect obstacles like walls and human beings in the path. It stops its way without hitting an obstacle.



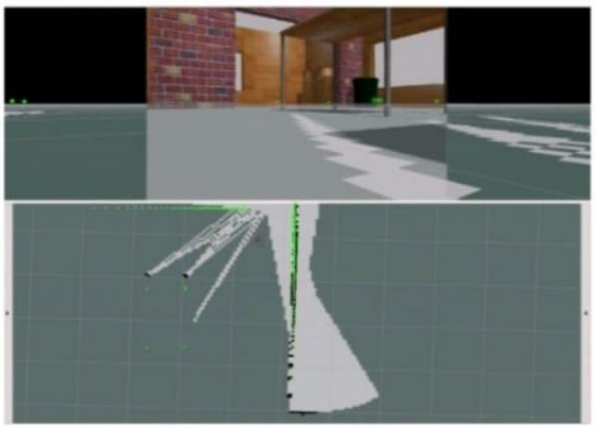
Ultrasonic Sensor

(c) ZED 2 stereo camera:

The ZED 2 camera offers greater depth detection with a 120° wide-angle field of view. The aim here is to combine with object detection, in particular human beings in movement. It can also perform multiple operations like Simultaneous localization and mapping Floor plane detection and Skeleton tracking. Robot view of the physical environment.



ZED 2 stereo camera



Robot view of physical environment

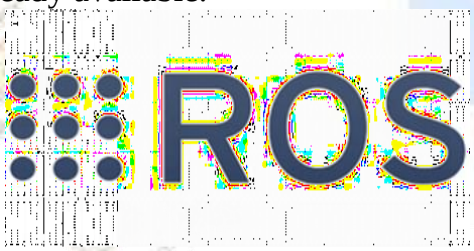


Python

B.Software Components:

(a) Robot Operating System(ROS):

The Robot Operating System (ROS) is an open-source framework that helps researchers and developers build and reuse code between robotics applications. ROS is also a global open-source community of engineers, developers and hobbyists who contribute to making robots better, more accessible, and available to everyone. With the ROS framework, new users can easily reuse code and develop robot applications. It is used in Robotics research. Since ROS is open source, all the needed communication tools and code are already available.



Robot Operating System

(b) Python:

Python is the language that is used with the Robot Operating System. This makes it highly relevant to robotics because you can use a Robot Operating System to control a robot. With Python programming language and Visual Components API, you are given a good platform for teaching, automating and post-processing robot programs.

Applications:

In a variety of industries, delivery robots have the potential to increase productivity, lower costs, and increase safety. Robots never have to isolate their thoughts between a huge number of things. They won't have sudden emergencies, and they won't be moved to wrap up an interchange time-sensitive task. They're by and large there, and they're doing what they ought to do. Computerization is frequently more dependable than human work. Due to their autonomy, delivery robots interact with the public primarily on their
own, without the help of a human operator. An integrated chain of custody system improves the accountability of how medications are delivered. By eliminating the need for human transportation and lowering the risk of contamination. Robots Assist in Addressing the Health care Labor Shortage. Using delivery robots can cut operating costs for hospitals. Accurate tracking of robot and equipment ensures a reduction in lost items. The time spent tracking down and refilling missed things is dismissed. Delivery and service robots can be used by senior citizens, people with disabilities, and individuals with limited mobility to complete tasks like obtaining scan reports and medications from pharmacies. Lastly, customers, businesses, and government organizations can benefit from the convenience and cost savings that delivery and service robots can offer.

Future Updations:

In the upcoming years, it is likely that the use of delivery robots will increase as technology develops and regulations become more enabling. From routine deliveries to more intricate, high-value applications, The autonomous machine space will move into many intriguing areas. Transmission Robots can use elevators and staircases to carry out deliveries on various floors and Increase the robot speed and weight loads of robot as possible. Robots will speed up and significantly secure the process. This method makes it easier to keep track of worker output and productivity. Robotic delivery is anticipated to improve in terms of quality, efficiency, accuracy, and safety.

Conclusion:

In conclusion, all the papers are reviewed in this paper and all the issues based on robots in delivery services such as poor quality of live tracking, prevention of theft or tampering of items as reports of a particular person are collected will be rectified in our model. Though the prototype mentioned in these papers are provided with some issues they provide a great insight about the delivery service robot. Further research is important in providing accuracy and efficiency in the working of the robot. By implementing the recommendations provided in all these papers, it is possible to increase the efficient working of autonomous delivery robots.

Acknowledgments:

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